

Independent Adversarial Audit Record

ERDOZ81 — Paper I — Quantitative Bound Gate

Audit target: **ERDOZ81-P1-BOUND-01**

Audited objective. Determine, from the supplied bundle alone, whether the chain proves that for every admissible split-neighborhood profile with clique size p and $q = |I_{\geq 2}|$, $M(\kappa) \geq (2p^2 - 2pq - q^2)/12 - p/2$.

Auditor stance. Adversarial. The chain was reconstructed from scratch and every load-bearing identity, inequality, regime split, and boundary case was re-derived by hand, re-checked symbolically in exact arithmetic, and stress-tested against an independent brute-force linear program that bypasses the paper's orbit machinery entirely.

1. Package-Completeness Result

Result: PACKAGE_COMPLETE. All 15 mandatory artifacts are present, readable, untruncated, and byte-size + SHA-256 matched against PACKAGE_MANIFEST.md. No substitution of instruction-stated formulas for proof artifacts was required.

#	Relative path	Bytes	SHA-256 (matched)
1	00_CONTROL/DEFINITION_REGISTRY.md	2933	a66c6f46...e108bfc1a
2	10_GATES/...3ORBIT-01/GATE_RECORD.md	3333	696babf2...b4b3d600ee6
3	10_GATES/...3ORBIT-SOLVE-01/GATE_RECORD.md	3168	be1ab939...3b13efc1ea
4	10_GATES/...BOUND-01/GATE_RECORD.md	2133	8de21e65...35dbddd81e1
5	10_GATES/...CONCAVE-01/GATE_RECORD.md	2474	53b6ed18...8054b15a
6	10_GATES/...PROFILE-01/GATE_RECORD.md	2786	a0ff77c8...403ebe6bd
7	10_GATES/...Q0-01/GATE_RECORD.md	2150	12296de5...74a9fa42bc
8	10_GATES/...SYMMETRY-01/GATE_RECORD.md	2551	ed00a671...73d6f01401
9	20_EVIDENCE/...EXACT_ORBIT_PROGRAM_PROOF_v0.1.md	5838	e09ac259...d8e9ec1a4
10	20_EVIDENCE/...EXACT_ORBIT_SOLUTION_PROOF_v0.1.md	5967	f44f1df8...a46e813b167
11	20_EVIDENCE/...QUANTITATIVE_BOUND_PROOF_v0.1.md	3289	dec319cf...30b270ecb5
12	20_EVIDENCE/...FIXED_POLYTOPE_CONCAVITY_PROOF_v0.1.md	1538	03f6e7da...c15a3c5653
13	20_EVIDENCE/...AFFINE_PROFILE_REDUCTION_PROOF_v0.1.md	1645	14f5d415...98b082c6bfd
14	20_EVIDENCE/...Q0_BRANCH_PROOF_v0.1.md	2197	abc721b1...8bdd389a4f6b
15	20_EVIDENCE/...PURE_PROFILE_SYMMETRIZATION_PROOF_v0.1.md	2923	5023b290...05cd333888db

2. Independence Disclosure

The audit used **only** the supplied bundle. No prior verdict, exploratory history, superseded archival material, or publication manuscript was consulted or relied upon as mathematical authority. The gate records' own pre-audit statuses (e.g. PROVISIONAL_PASS, "evidence CANDIDATE", self-audit events) were read as claims to be tested, not as evidence. Independent tooling: hand re-derivation; SymPy exact-symbolic identity checking; and an independently coded exact-rational / HiGHS linear-program solver over the full triangle-cover polytope P_p that does not use the orbit reduction, symmetrization, or candidate values U , D , H .

3. Claim-by-Claim Findings (Load-Bearing Chain)

3.1 Definitions and domains (DEFINITION_REGISTRY / DEF-01)

$M(\kappa) := M_1(\kappa) = \min_{z \in P} \langle 1-\kappa, z \rangle$ with $P_p = \{z \in [0,1]^{E(K_p)} : z_{e1} + z_{e2} + z_{e3} \geq 1 \text{ for every triangle}\}$. For $p < 3$ there are no triangle inequalities, so $P_p = [0,1]^E$ ($P_0, P_1 =$ the empty vector). $p=|K|, q=|I|_{\geq 2}$, and $a_e^S = 1_{\{e \subseteq S\}} / (|S|-1)$ for $|S| \geq 2$. Definitions are self-consistent; the min exists (nonempty compact polytope); denominators are guarded ($|S| \geq 2$); degenerate regimes $p \leq 2$ and empty orbits are explicitly covered. No sign condition is imposed on the objective vector, which the downstream negative-coefficient handling requires. **VERIFIED.**

3.2 Fixed-polytope concavity (CONCAVE-01)

$\Phi(c) = \min_{x \in P} \langle c, x \rangle$ on a fixed nonempty P satisfies $\Phi(\sum \lambda_i c^{(i)}) \geq \sum \lambda_i \Phi(c^{(i)}) \geq \min_i \Phi(c^{(i)})$ for a convex combination. The proof is the textbook pointwise-minimum-of-affine argument; inequality direction is correct (concavity, not convexity). Crucially the feasible set is held fixed and only the objective vector varies — the exact hypothesis the profile step needs. **VERIFIED.**

3.3 Pure-profile reduction (PROFILE-01) — requires $q > 0$

$\kappa = \sum_S n_S a^S$ holds coordinatewise (each active vertex has one exact type $S=N(v)$). With $\lambda_S = n_S/q$ (≥ 0 , summing to 1) and $\kappa^S = q \cdot a^S$, one gets $\kappa = \sum_S \lambda_S \kappa^S$. Applying 3.2 to the fixed polytope with the objective vectors $b - \kappa^S$: $M_b(\kappa) \geq \sum_S \lambda_S M_b(\kappa^S) \geq \min_S M_b(\kappa^S)$. Convex combination is on objective vectors, polytope fixed — consistent with 3.2's hypothesis. Independently confirmed on 2000 random mixed profiles (dominance never violated). **VERIFIED.**

3.4 Symmetrization (SYMMETRY-01)

Under $\Gamma = \text{Sym}(S) \times \text{Sym}(K \setminus S)$: the box and triangle constraints are permutation-invariant ($g \cdot P_p = P_p$); the objective coefficient is constant on each edge class $SS / SO / OO$, so $\langle c, gz \rangle = \langle c, z \rangle$ even when the SS coefficient $1-q/(s-1)$ is negative; averaging an optimizer over Γ stays feasible (convexity) and optimal, and is constant on each nonempty orbit. Negative-coefficient case explicitly and correctly handled. **VERIFIED.**

3.5 Orbit-program equivalence (3ORBIT-01)

Edge orbits SS, SO, OO and the four triangle types SSS ($3\alpha \geq 1$, exists $s \geq 3$), SSO ($\alpha + 2\beta \geq 1, s \geq 2, o \geq 1$), SOO ($2\beta + \gamma \geq 1, o \geq 2$), OOO ($3\gamma \geq 1, o \geq 3$) are exhaustive and their existence conditions correct. Both directions of equivalence hold and objective multiplicities $C(s,2)$, so, $C(o,2)$ are correct, giving $A = s(s-1-q)/2, B = so, C = o(o-1)/2$. Constraints for nonexistent triangle types are omitted per regime (notably no $3\gamma \geq 1$ at $o=2$). **VERIFIED.**

3.6 Exact orbit solution (3ORBIT-SOLVE-01)

Solved by fixing α , minimizing exactly over (β, γ) , then minimizing the resulting affine function of α on $[\blacksquare, 1]$ ($\blacksquare=0$ if $s=2$, else $1/3$) — a valid full-polytope argument, not a vertex guess. The sign switch is governed exactly by $B - 2C = o(s-o-1)$. Result: $\min F = \min\{U, D\}$ for $0 \leq o \leq 2$ and $\min\{U, D, H\}$ for $o \geq 3$, with $U = (A+B+C)/3, D = A+C, H = A+(B+C)/3$. The $s=2$ lower bound $\alpha \geq 0$ adds no new optimum (monotonicity checked). I reproduced each regime by hand and, independently, an exact/HiGHS LP over the entire polytope reproduced $M(\kappa^S)$ to the envelope in all 728 pure-profile cases tested ($p \leq 8, q$ up to $4p+2$). **VERIFIED.**

3.7 The $q=0$ branch (Q0-01)

$q=0 \Rightarrow I_{\geq 2} = \emptyset, \kappa=0$, and every triangle lies in K (I is independent). For $p \leq 2, M(0)=0$ and $R(p,0) - p/2 = (p^2 - 3p)/6 \leq 0$. For $p \geq 3, z_e = 1/3$ is feasible and, since each edge is in exactly $p-2$ triangles, summing all triangle constraints forces optimality, giving $M(0) = C(p,3)/(p-2) = p(p-1)/6$, so $M(0) - R(p,0) = -p/6 \geq -p/2$. No division by q occurs. Exact LP confirms $M(0) = p(p-1)/6$ for $p=3..9$. **VERIFIED.**

3.8 Quantitative envelope (BOUND-01)

Target $R = (2p^2 - 2pq - q^2)/12$ is the quantity fixed by the audit objective; the proof's §1 “internal derivation” merely shows R is forced by a downstream normalization and is not needed to establish the inequality itself. The three candidate

identities were verified *exactly* in symbolic algebra (residual 0 in every case):

- $12(U-R) = q(2p+q-2s)-2p = q(2o+q)-2p \geq -2p \Rightarrow U-R \geq -p/6.$
- $12(D-R) = 12o^2-6o(2p-q)+(2p-q)^2-6p$; and $12u^2-6uv+v^2=12(u-v/4)^2+v^2/4 \geq 0 \Rightarrow D-R \geq -p/2.$
- $12(H-R) = (2s-q)^2+2q(p-s)-2p-4s \geq -2p-4s \geq -6p$ (using $s \leq p$) $\Rightarrow H-R \geq -p/2.$

Since R depends only on (p,q) and every candidate satisfies candidate $\geq R-p/2$, the envelope $\min\{U,D,H\} \geq R-p/2$, hence $M(\kappa^S) \geq R-p/2$ for every pure profile, and by 3.3 $M(\kappa) \geq R-p/2$ for every profile with $q>0$. Combined with 3.7 the bound holds for all admissible parameters. **VERIFIED.**

4. Independent Derivations and Falsification Attempts

(a) R is forced. Solving $C(p,2)-2R+p=(p+q)^2/6+p/2$ gives exactly $R=(2p^2-2pq-q^2)/12$ (SymPy: residual 0).

(b) All four algebraic identities (R, U, D, H) and the quadratic-form decomposition were confirmed with residual 0 in exact symbolic arithmetic. $B-2C$ factors exactly as $o(s-o+1)$; $H-D=(B-2C)/3$, matching the tie analysis.

(c) Full-polytope LP (bypasses the orbit machinery). An independently coded solver over P_p was run on 728 pure profiles ($p=2..8$; every s ; $q=1..4p+2$). In every case the true LP optimum equalled the claimed envelope and satisfied $M \geq R-p/2$. Zero failures.

(d) Mixed-profile stress test. 2000 random neighborhood mixtures ($p=3..7$) all satisfied both $M \geq R-p/2$ and pure-profile dominance. Zero failures.

(e) Exact-rational corroboration. Vertex-enumeration LP in exact fractions ($p \leq 4$ and full $p=3$ sweep) reproduced all values with no floating-point ambiguity; $q=0$ gave $M(0)=p(p-1)/6$ exactly.

(f) Tightness / mis-statement test. Equality $M=R-p/2$ is *achieved* at ($s=p$, $o=0$, $q=2p$): exact LP gives $M=-3,-6,-10$ for $p=2,3,4$, matching $R-p/2$ exactly. The bound is therefore the exact envelope, not loose — strong evidence the target expression is correctly stated and cannot be strengthened.

(g) Attempts to break it. Large q (well beyond $2p$, making the SS coefficient strongly negative and $A \leq 0$), the $s=2$ degenerate branch, $o=0/1/2 \geq 3$ transitions, and $B=2C$ ties were all probed; no feasible point below $R-p/2$ and no omitted orbit optimum was found.

5. Dependency Table

Dependency / Claim	Status
DEF-01 — definitions, domains, degenerate regimes	VERIFIED
CONCAVE-01 — fixed-polytope concavity, inequality direction	VERIFIED
PROFILE-01 — affine profile identity + pure-profile dominance ($q>0$)	VERIFIED
SYMMETRY-01 — stabilizer averaging incl. negative coefficients	VERIFIED
3ORBIT-01 — orbit-program equivalence, both directions	VERIFIED
3ORBIT-SOLVE-01 — exact orbit optimum in all regimes	VERIFIED
Q0-01 — $q=0$ branch and small p	VERIFIED
BOUND-01 C-BOUND-01 — internal formula for $R(p,q)$	VERIFIED
BOUND-01 C-BOUND-02 — $U-R$ identity and bound	VERIFIED
BOUND-01 C-BOUND-03 — $D-R \geq -p/2$	VERIFIED
BOUND-01 C-BOUND-04 — $H-R \geq -p/2$ ($o \geq 3$)	VERIFIED
BOUND-01 C-BOUND-05 — $q=0$ branch	VERIFIED
BOUND-01 C-BOUND-06 — global envelope $M(\kappa) \geq R-p/2$	VERIFIED
Downstream graph identity $ E -2v_{con}=\dots$ (motivation for R; OUT OF SCOPE per §8)	UNCHECKED

No dependency is marked FAILED. The single UNCHECKED item is explicitly outside this gate's scope (§8) and, as noted, is not load-bearing for the inequality under audit.

6. Material Objections, Minor Observations, Required Actions

Material objections: None. No incorrect identity, missing regime, reversed inequality, unstated hypothesis, non-exhaustive optimization, or dependency mismatch was found in the in-scope chain.

Minor observations (non-blocking):

- The §1 derivation of R invokes a downstream graph-theoretic identity ($|E(G)| - 2v_{\text{con}} = C(p, 2) - 2M(\kappa) + b_1$) that is asserted, not proved, in the bundle. This is motivation only; the audited inequality uses R as given and does not depend on that identity. It falls under the excluded assembly/transfer scope (§8).
- Gate-record bookkeeping is internally inconsistent: several T2 gates read “Mathematical status: PASS / dependency health CLEAN” while simultaneously stating “Admitted Evidence: None; evidence remains CANDIDATE”, and BOUND-01 is PROVISIONAL_PASS with UNVERIFIED health. These are administrative metadata discrepancies, not mathematical defects.
- DEF-01, PHASE1-01, and LIT-01 are referenced but not all supplied as standalone gate records. DEF content is supplied via the registry and is self-contained; PHASE1-01 is only an assembly hook for Q0's empty-support statement (whose math is independently verified); LIT-01 is CITATION_ONLY and the proof uses no external theorem.

Required actions: None to promote this gate on its stated claim. For the eventual full theorem (out of scope here), the downstream identity of §1 and the T4 assembly should be audited separately before end-to-end certification.

7. Final Verdict

RECOMMEND_PASS

The package is complete and every load-bearing link of ERDOZ81-P1-BOUND-01 — definitions and domains, fixed-polytope concavity, pure-profile reduction, symmetrization, orbit-program equivalence, the exact orbit solution, the $q=0$ branch, and the quantitative envelope — was independently verified without material objection. The inequality $M(\kappa) \geq (2p^2 - 2pq - q^2)/12 - p/2$ holds for every admissible parameter set, was re-derived by hand, confirmed in exact symbolic algebra, corroborated by an independent full-polytope LP that does not use the paper's reduction, and shown to be tight. This verdict certifies only ERDOZ81-P1-BOUND-01 and its supplied dependencies; per §8 it does not certify the finite-theorem assembly, any integral transfer, publication novelty, or the T4 Assembly Gate.